

# DATASHEET

## Variable Speed Drives



Product coding : CFW500A06P1T4NB20  
 Product code : 11575841  
 Product reference : CFW500  
 Accessory module (control) : CFW500-IOS

### Basic data

Power supply : 380-480 V  
 Input minimum-maximum voltage : 323-528 V  
 - In : 3  
 - Out : 3

| Supply voltage range             | 380-480 V            |                     |
|----------------------------------|----------------------|---------------------|
| Overload cycle                   | Normal Overload (ND) | Heavy Overload (HD) |
| Rated current (HD)               | 6.1                  | 6.1                 |
| Overload current for 60 sec (HD) | 9                    | 9                   |
| Overload current for 3 sec (HD)  | 10                   | 10                  |

Maximum applicable motor:

| Voltage/Frequency | Power (HP/kW) [1]    |                     |
|-------------------|----------------------|---------------------|
|                   | Normal Overload (ND) | Heavy Overload (HD) |
| 380V / 50Hz       | Not applicable       | 3 / 2,2             |
| 380V / 60Hz       | Not applicable       | 3 / 2,2             |
| 400V / 50Hz       | Not applicable       | 4 / 3               |
| 400V / 60Hz       | Not applicable       | 3 / 2,2             |
| 440V / 50Hz       | Not applicable       | 4 / 3               |
| 440V / 60Hz       | Not applicable       | 4 / 3               |
| 460V / 60Hz       | Not applicable       | 4 / 3               |
| 480V / 60Hz       | Not applicable       | 4 / 3               |

Accessory module (control) : CFW500-IOS  
 Dynamic braking [2] : Standard without braking  
 External electronic supply 24Vcc : Not available  
 Safety Stop : Not available  
 Internal RFI filter : Without filter  
 External RFI filter : Not available  
 Link Inductor : No  
 Memory card : Not included in the product  
 USB port : Only with plug-in  
 Line frequency : 50/60Hz  
 Line frequency range (minimum - maximum) : 48-62 Hz  
 Phase unbalance : less or equal to 3% of input rated line voltage  
 Transient voltage and overvoltage : Category III  
 Single-phase input current [3] : Not applicable  
 Three-phase input current [3] : 7,4 A  
 Power factor : 0,83  
 Displacement factor : 0,98  
 Rated efficiency : ≥ 97%  
 Maximum connections (power up cycles - on/off) per hour : 10 (1 each 6 minutes)  
 DC power supply : Not allow  
 Standard switching frequency : 5 kHz  
 Selectable switching frequency : 2,5 and 15 kHz  
 Real-time clock : Not available  
 COPY Function : Yes, by MMF  
 Dissipated power:

| Mounting type | Overload       |                |
|---------------|----------------|----------------|
|               | ND             | HD             |
| Surface       | 105 W          | 105 W          |
| Flange        | Not applicable | Not applicable |

Output voltage : 24 Vcc  
 Maximum capacity : 150 mA  
 Power supply : Switched-mode power supply  
 Control method : V/f, VVW, Sensorless and Encoder  
 Encoder interface : Only with plug-in  
 Control output frequency : 0-500 Hz  
 Frequency resolution : 0,015 Hz  
 - Speed resolution : 1% of rated speed  
 - Speed range : 1:20

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|                    |                         |
|--------------------|-------------------------|
| - Speed resolution | : 1% of rated speed     |
| - Speed range      | : 1:30                  |
| - Speed resolution | : 0,5% of rated speed   |
| - Speed range      | : 1:100                 |
| - Speed resolution | : 0,1% of nominal speed |
| - Speed range      | : Up to 0 rpm           |

|                             |                            |
|-----------------------------|----------------------------|
| Quantity (standard)         | : 1                        |
| Levels                      | : 0-10V, 0-20mA and 4-20mA |
| Impedance for voltage input | : 100 kΩ                   |
| Impedance for current input | : 500 Ω                    |
| Function                    | : Programmable             |
| Maximum allowed voltage     | : 30 Vcc                   |

|                         |                           |
|-------------------------|---------------------------|
| Quantity (standard)     | : 1                       |
| Activation              | : Active low and high     |
| Maximum low level       | : 5 V (low) e 15 V (high) |
| Minimum high level      | : 9 V (low) e 20 V (high) |
| Input current           | : 4,5 mA                  |
| Maximum input current   | : 5,5 mA                  |
| Function                | : Programmable            |
| Maximum allowed voltage | : 30 Vcc                  |

### Analog outputs

|                                        |                                     |
|----------------------------------------|-------------------------------------|
| Analogic outputs - Quantity (standard) | : 1                                 |
| Levels                                 | : 0 to 10V, 0 to 20mA and 4 to 20mA |
| RL for voltage output                  | : 10 kΩ                             |
| RL for current output                  | : 500 Ω                             |
| Function                               | : Programmable                      |

|                                       |                                  |
|---------------------------------------|----------------------------------|
| Digital outputs - Quantity (standard) | : 1 NO/NC relay and 1 transistor |
| Maximum voltage                       | : 240 Vca and 24 Vcc             |
| Maximum current                       | : 0,5 A and 150 mA               |
| Function                              | : Programmable                   |

- Modbus-RTU (with accessory: Any plug-in module)
- Modbus/TCP (with accessory CFW500-CEMB-TCP)
- Profibus DP (with accessory: CFW500-CPDP)
- Profibus DPV1 (with accessory: CFW500-CPDP)
- Profinet (with accessory CFW500-CEPN-IO)
- CANopen (with accessory: CFW500-CCAN)
- DeviceNet (with accessory: CFW500-CCAN)
- EtherNet/IP (with accessory CFW500-CETH-IP)
- EtherCAT (Not available)
- BACnet (Not aplicable)

- Output phase-phase overcurrente/Short
- Overcurrent/Short circuit phase-ground
- Under/Overvoltage in power
- Heat sink overtemperature
- Motor overload
- IGBT's modules overload
- Fault/External alarm
- Programming error

|                                   |                           |
|-----------------------------------|---------------------------|
| Avaliability                      | : Included in the product |
| Installation                      | : Fixed HMI               |
| Number of HMI buttons             | : 9                       |
| Display                           | : Numeric LCD             |
| Indication accuracy               | : 5% of rated current     |
| Speed resolution                  | : 0,1 Hz                  |
| Standard HMI degree of protection | : IP20                    |
| HMI battery type                  | : Not applicable          |
| HMI battery life expectancy       | : Not applicable          |
| Remote HMI type                   | : Accessory               |
| Remote HMI frame                  | : Not applicable          |
| Remote HMI degree of protection   | : IP54                    |

|                     |        |
|---------------------|--------|
| Enclosure           | : IP20 |
| Degree of pollution | : 2    |

|                   |       |
|-------------------|-------|
| RoHS              | : Yes |
| Conformal Coating | : 3C2 |

|          |                   |
|----------|-------------------|
| - Size   | : A               |
| - Height | : 189 mm / 7.4 in |
| - Width  | : 75 mm / 2.95 in |

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- Depth : 150 mm / 5.91 in  
 - Weight : 0,8 kg / 1.8 lb

### Mechanical Installation

Mounting position : Surface or DIN rail  
 Fixing screw : M4  
 Tightening torque : 2 N.m / 1.48 lb.ft  
 Allows side-by-side assembly : Yes, maximum ambient temperature 40°C  
 - Top : 15 mm / 0.59 in  
 - Bottom : 40 mm / 1.57 in  
 - Front : 30 mm / 1.18 in  
 - Side : 10 mm / 0.39 in

Cable gauges and tightening torques:

|  |                                           |                      |
|--|-------------------------------------------|----------------------|
|  | 1,5 mm <sup>2</sup> (16 AWG)              | 0,5 N.m / 0.37 lb.ft |
|  | Not applicable                            | 0,5 N.m / 0.37 lb.ft |
|  | 2,5 mm <sup>2</sup> (14 AWG)              | 0,5 N.m / 0.37 lb.ft |
|  | 0,5 to 1,5 mm <sup>2</sup> (20 to 14 AWG) | 0,5 N.m / 0.37 lb.ft |

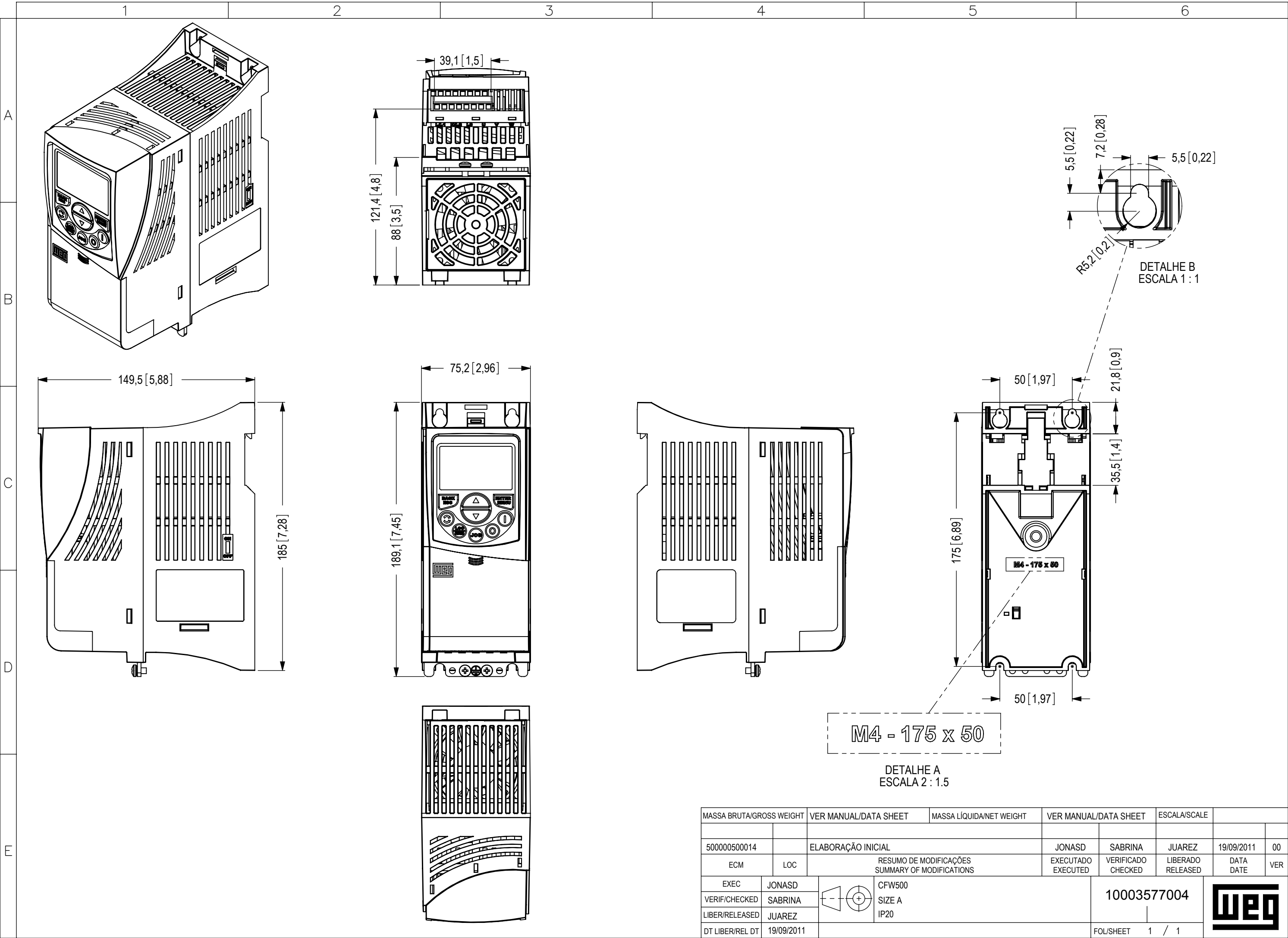
SoftPLC : Yes, incorporated  
 Maximum breaking current : Not available  
 Minimum resistance for the brake resistor : Not available  
 Recommended aR fuse : FNH00-20K-A  
 Recommended circuit breaker : MPW18-3-U010  
 Disconnect switch : Not applicable  
 Motor coupling box : Not applicable

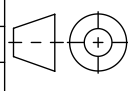
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|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | -<br>-<br>-<br>-<br>-<br>-                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|  | - EN 61800-3 - Adjustable speed electrical power drive systems - Part 3: EMC product standard including specific test methods.<br>- EN 55011 - Limits and methods of measurement of radio disturbance characteristics of industrial, scientific and medical (ISM) radio-frequency equipment.<br>- CISPR 11 - Industrial, scientific and medical (ISM) radio-frequency equipment<br>- Electromagnetic disturbance characteristics - Limits and methods of measurement.<br>- EN 61000-4-2 - Electromagnetic compatibility (EMC) - Part 4: Testing and measurement techniques - Section 2: Electrostatic discharge immunity test.<br>-<br>-<br>- |
|  | - EN 60529 e UL 50                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

### Certifications

### Notes

- 1) Motor power is orientative, valid for standard WEG Motors of IV poles. The correct sizing must be done according to the nominal current of the motor used, which must be less than or equal to the rated output current of the inverter;
- 2) Braking resistor is not included;
- 3) Considering minimum line impedance of 1%;
- 4) For more information, refer to the user manual of CFW500;
- 5) All images are merely illustrative.
- 6) For operation with switching frequency above nominal, apply derating to the output current (refer to the user manual).



|                          |  |                       |  |                                                                                                                |  |                       |  |                       |  |                                                                                       |  |
|--------------------------|--|-----------------------|--|----------------------------------------------------------------------------------------------------------------|--|-----------------------|--|-----------------------|--|---------------------------------------------------------------------------------------|--|
| MASSA BRUTA/GROSS WEIGHT |  | VER MANUAL/DATA SHEET |  | MASSA LÍQUIDA/NET WEIGHT                                                                                       |  | VER MANUAL/DATA SHEET |  | ESCALA/SCALE          |  |                                                                                       |  |
|                          |  |                       |  |                                                                                                                |  |                       |  |                       |  |                                                                                       |  |
| 500000500014             |  | ELABORAÇÃO INICIAL    |  |                                                                                                                |  | JONASD                |  | SABRINA               |  | JUAREZ                                                                                |  |
| ECM                      |  | LOC                   |  | RESUMO DE MODIFICAÇÕES<br>SUMMARY OF MODIFICATIONS                                                             |  |                       |  | EXECUTADO<br>EXECUTED |  | LIBERADO<br>RELEASED                                                                  |  |
|                          |  |                       |  |                                                                                                                |  |                       |  |                       |  | DATA<br>DATE                                                                          |  |
|                          |  |                       |  |                                                                                                                |  |                       |  |                       |  | VER                                                                                   |  |
| EXEC                     |  | JONASD                |  |  CFW500<br>SIZE A<br>IP20 |  |                       |  | 10003577004           |  |  |  |
| VERIF/CHECKED            |  | SABRINA               |  |                                                                                                                |  |                       |  |                       |  |                                                                                       |  |
| LIBER/RELEASED           |  | JUAREZ                |  |                                                                                                                |  |                       |  |                       |  |                                                                                       |  |
| DT LIBER/REL DT          |  | 19/09/2011            |  |                                                                                                                |  |                       |  | FOL/SHEET 1 / 1       |  |                                                                                       |  |